

User Management in Linux

Introduction to User Management in Linux

Linux is a multi-user operating system, meaning multiple users can operate on a system simultaneously. Proper user management ensures security, controlled access, and system integrity.

Key files involved in user management:

- `/etc/passwd` – Stores user account details.
- `/etc/shadow` – Stores encrypted user passwords.
- `/etc/group` – Stores group information.
- `/etc/gshadow` – Stores secure group details.

Creating Users in Linux

To create a new user in Linux, use:

useradd Command (For most Linux distributions)

```
useradd username
```

This creates a user without a home directory.

To create a user with a home directory:

```
useradd -m username
```

To specify a shell:

```
useradd -s /bin/bash username
```

adduser Command (For Debian-based systems)

```
adduser username
```

This is an interactive command that asks for a password and additional details.

Managing User Passwords

To set or change a user's password:

```
passwd username
```

Enforcing Password Policies

- **Password expiration:** Set password expiry days

```
chage -M 90 username
```

- **Lock a user account**

`passwd -l username`

- **Unlock a user account**

`passwd -u username`

Modifying Users

Modify an existing user with `usermod`:

- Change the username:

`usermod -l new_username old_username`

- Change the home directory:

`usermod -d /new/home/directory -m username`

- Change the default shell:

`usermod -s /bin/zsh username`

Deleting Users

To remove a user but keep their home directory:

`userdel username`

To remove a user and their home directory:

`userdel -r username`

Working with Groups

Creating Groups

`groupadd groupname`

Adding Users to Groups

`usermod -aG groupname username`

Viewing Group Memberships

`groups username`

Changing Primary Group

`usermod -g new_primary_group username`